

Homework Assignment 1A

Class-D Amplifier

ET4382 - Digital IC Design

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Part 1

LC Lowpass Filter Design

Part 1: Specifications & Design

Specifications:

Load: $R = 4 \text{ Ohm}$

Attenuation @ 1MHz: $> 50 \text{ dB}$

Quality factor: $0.7 < Q < 1.0$

E12 series components, $L \pm 10\%$, $C \pm 20\%$

Topology: 2nd-order LC lowpass

L in series, C in parallel with R (shunt to GND)

Formulas:

$$f_0 = 1 / (2\pi\sqrt{LC})$$

$$Q = R * \sqrt{C/L}$$

$$\text{Attenuation} (f \gg f_0) = 40 * \log_{10}(f/f_0) \text{ dB}$$

Selected E12 values: $L = 18 \text{ uH}$, $C = 820 \text{ nF}$

Part 1: Nominal Performance

Component values:

$L = 18 \text{ uH}$ (E12, +/-10%)

$C = 820 \text{ nF}$ (E12, +/-20%)

$R = 4 \text{ Ohm}$ (load)

Calculated performance:

$f_0 = 1/(2\pi\sqrt{18\text{u} * 820\text{n}}) = 41.4 \text{ kHz}$

$Q = 4 * \sqrt{820\text{n} / 18\text{u}} = 0.854$

Attenuation @ 1MHz = $40 * \log_{10}(1\text{e}6/41430) = 55.3 \text{ dB}$

Simulated (Cadence AC analysis):

$f_{-3\text{dB}} = 48.66 \text{ kHz}$

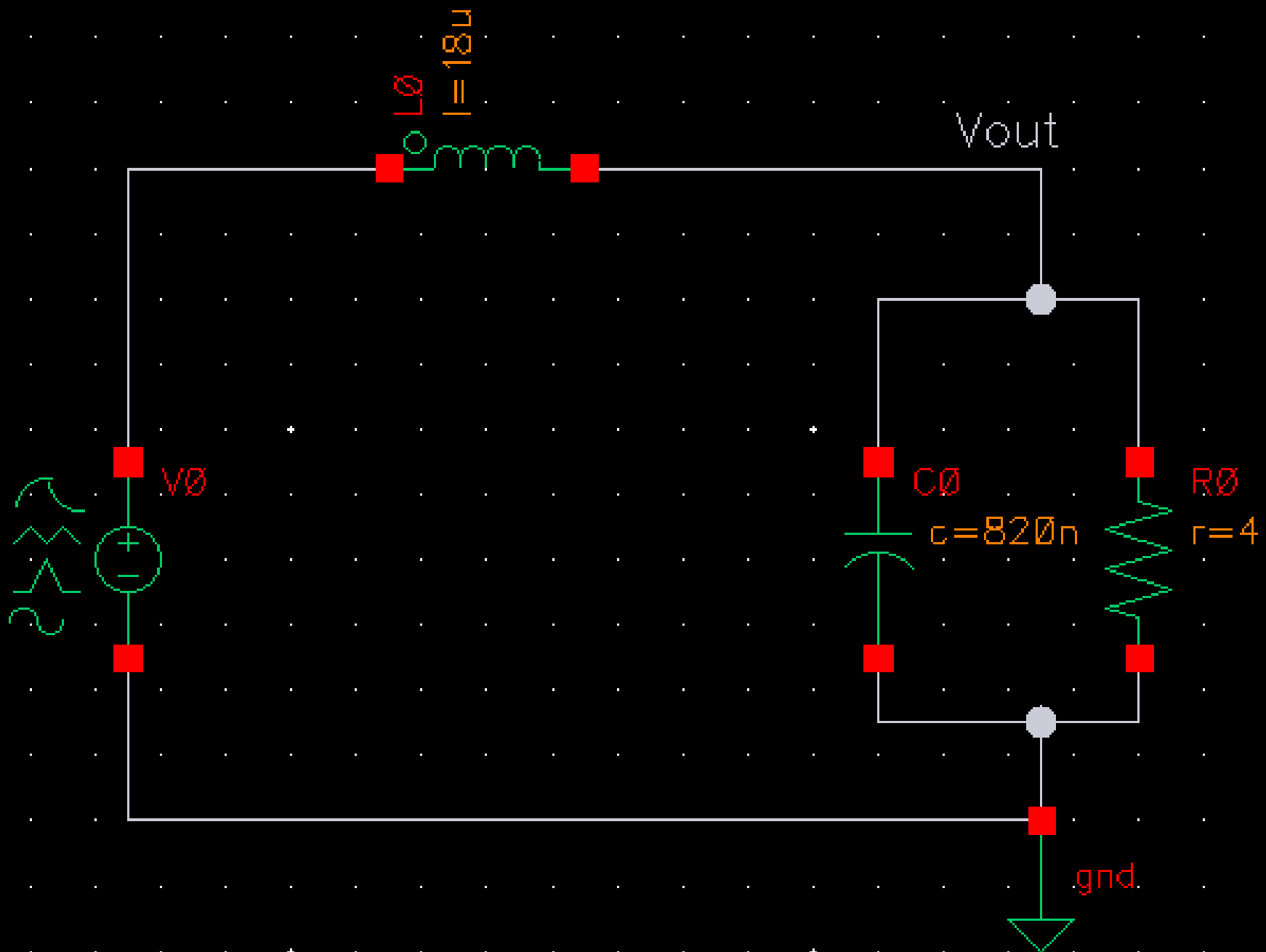
Attenuation @ 1MHz = 55.45 dB

Q confirmed by Bode shape (no peaking)

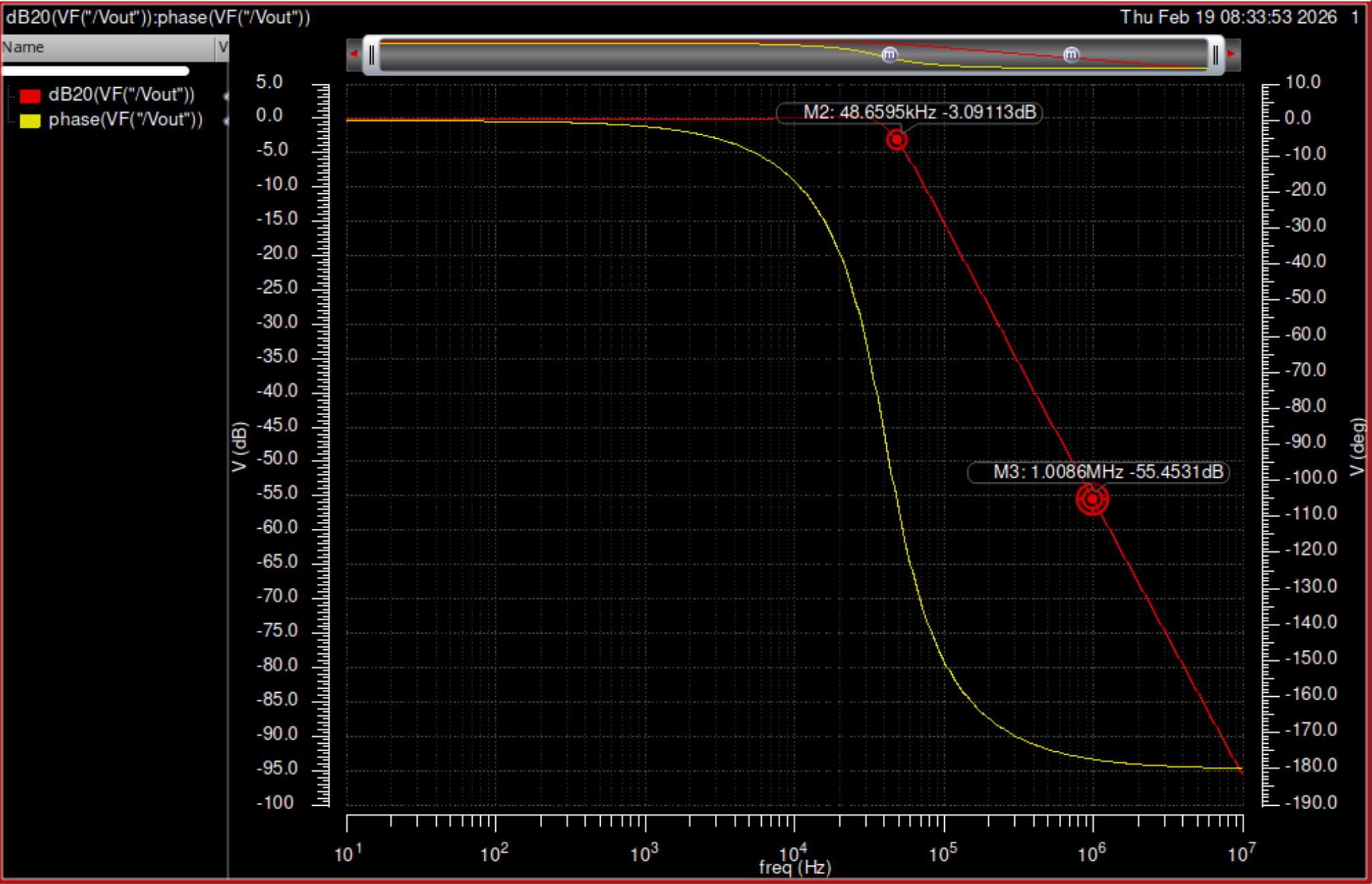
Part 1: Worst-Case Tolerance Analysis

Corner	L	C	f0 (kHz)	Q	Atten @1MHz (dB)
Nominal	18 uH	820 nF	41.4	0.854	55.3
L-10% C-20%	16.2 uH	656 nF	48.8	0.986	52.4
L+10% C+20%	19.8 uH	984 nF	36.2	0.728	58.9
L-10% C+20%	16.2 uH	984 nF	39.9	0.986	55.0
L+10% C-20%	19.8 uH	656 nF	44.1	0.728	53.1
All corners				0.73-0.99	>52.4

Part 1: Schematic



Part 1: Bode Plot



AC analysis: magnitude & phase, markers at f_{-3dB} and 1MHz

Part 2

Double-Sided PWM Generator

Part 2: Circuit Description

Components:

Comparator (ahdlLib): sigout_high = +10V, sigout_low = -10V

Triangle carrier (vpulse): V1=-1, V2=+1, Rise=499n, Fall=499n, Period=1us

Sine input (vsin): variable amplitude and frequency

LC filter + 4 Ohm load (same as Part 1)

Double-sided (naturally-sampled) PWM:

Input sine compared with triangle carrier at $f_{\text{PWM}} = 1 \text{ MHz}$

Output switches between +10V and -10V

LC filter recovers the audio-frequency content

Simulation: transient 5ms, conservative accuracy

Spectrum: dB20(dft(VT("/Vout") 1m 5m 4096))

Part 2: Simulation Results

1 kHz, 0.1 FS:

Clean $\sim \pm 1V$ sine at output, strong 1kHz spectral peak

1.001 MHz (fPWM + 1kHz), 0.1 FS:

Almost no output - LC filter attenuates carrier-frequency content by >50dB

2.001 MHz (2*fPWM + 1kHz), 0.1 FS:

A 1kHz sine appears! Signal aliases back through double-sided PWM

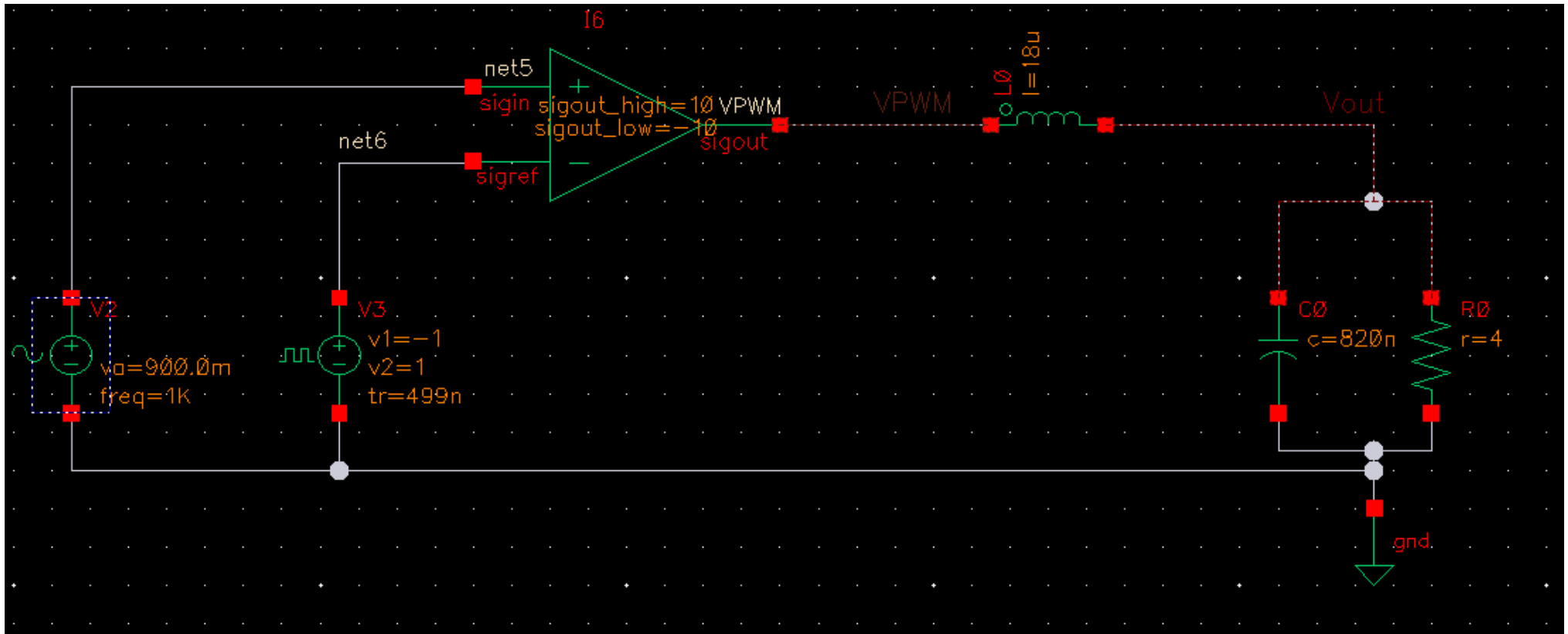
Sidebands around 2*fPWM fold into baseband - key PWM property

1 kHz, 0.9 FS:

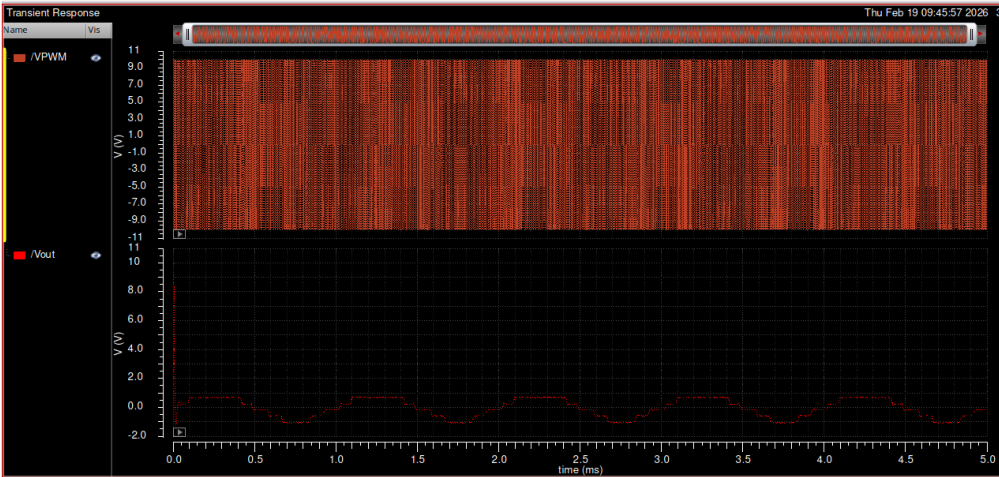
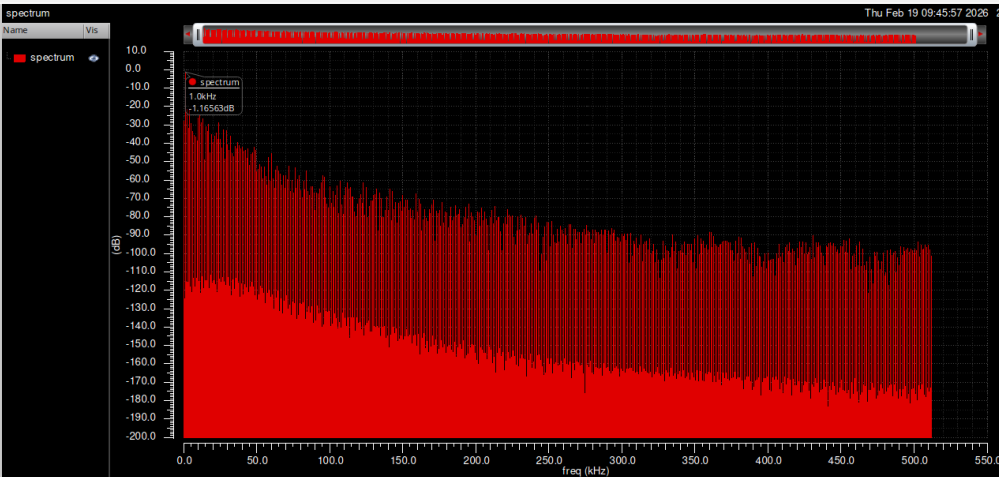
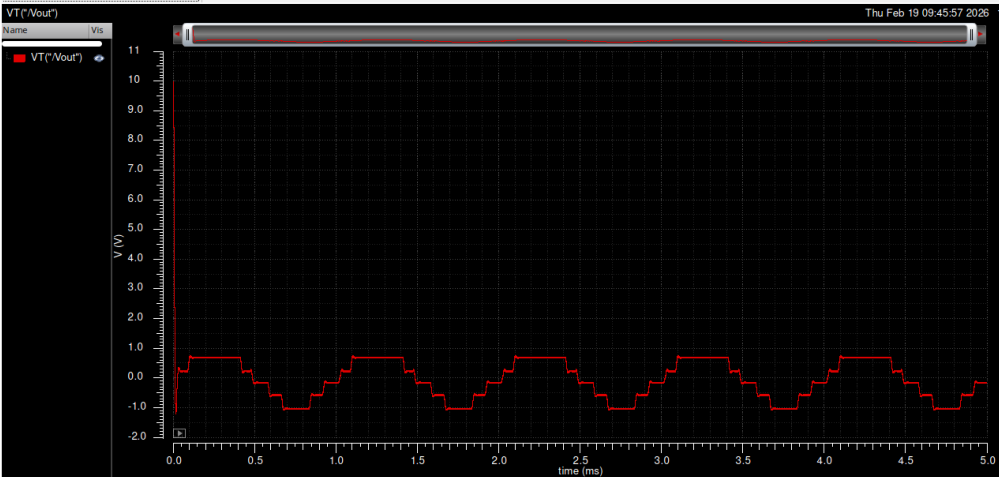
Large $\sim \pm 9V$ sine wave, near full modulation

PWM pulse widths vary from very narrow to very wide

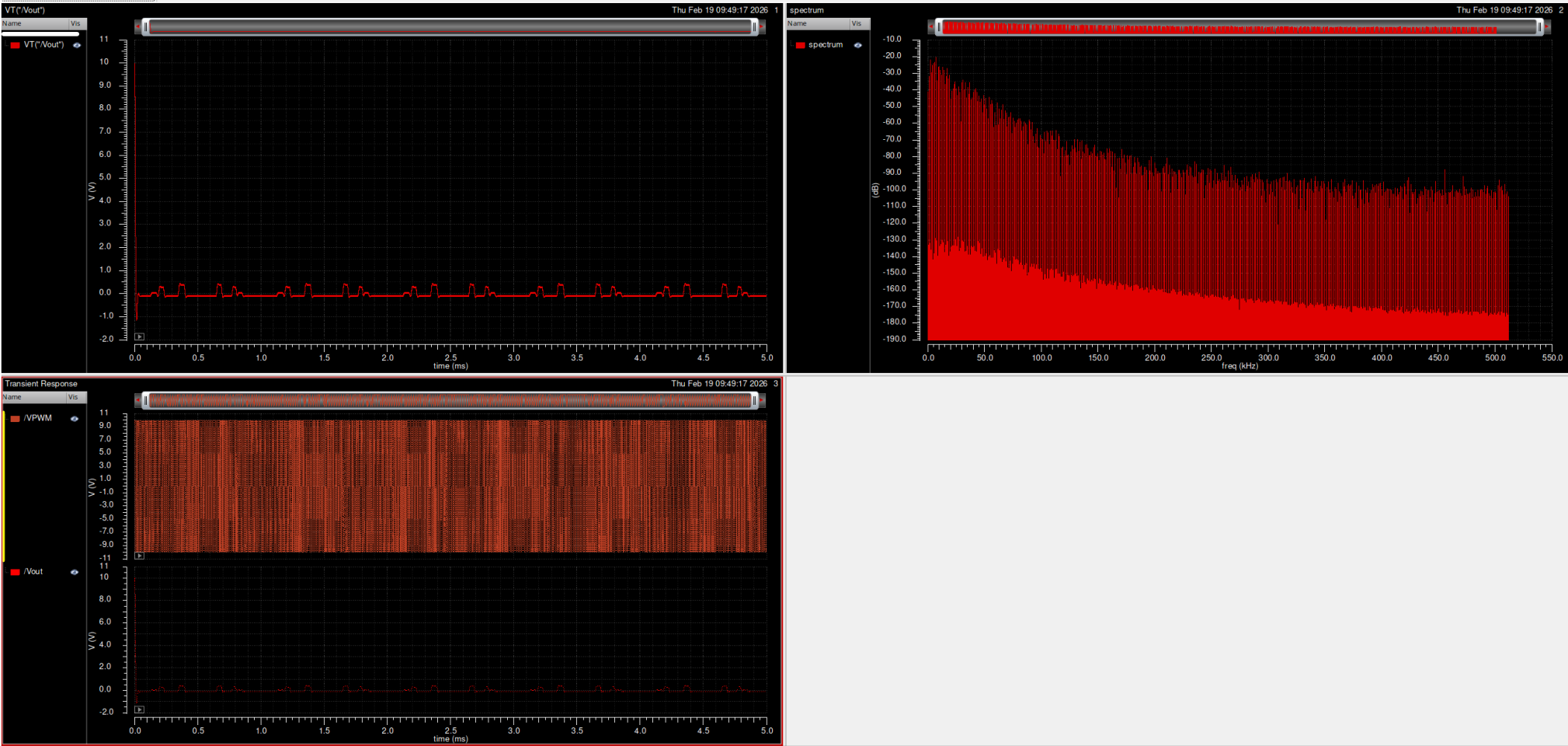
Part 2: Schematic



Part 2: 1kHz, 0.1 FS

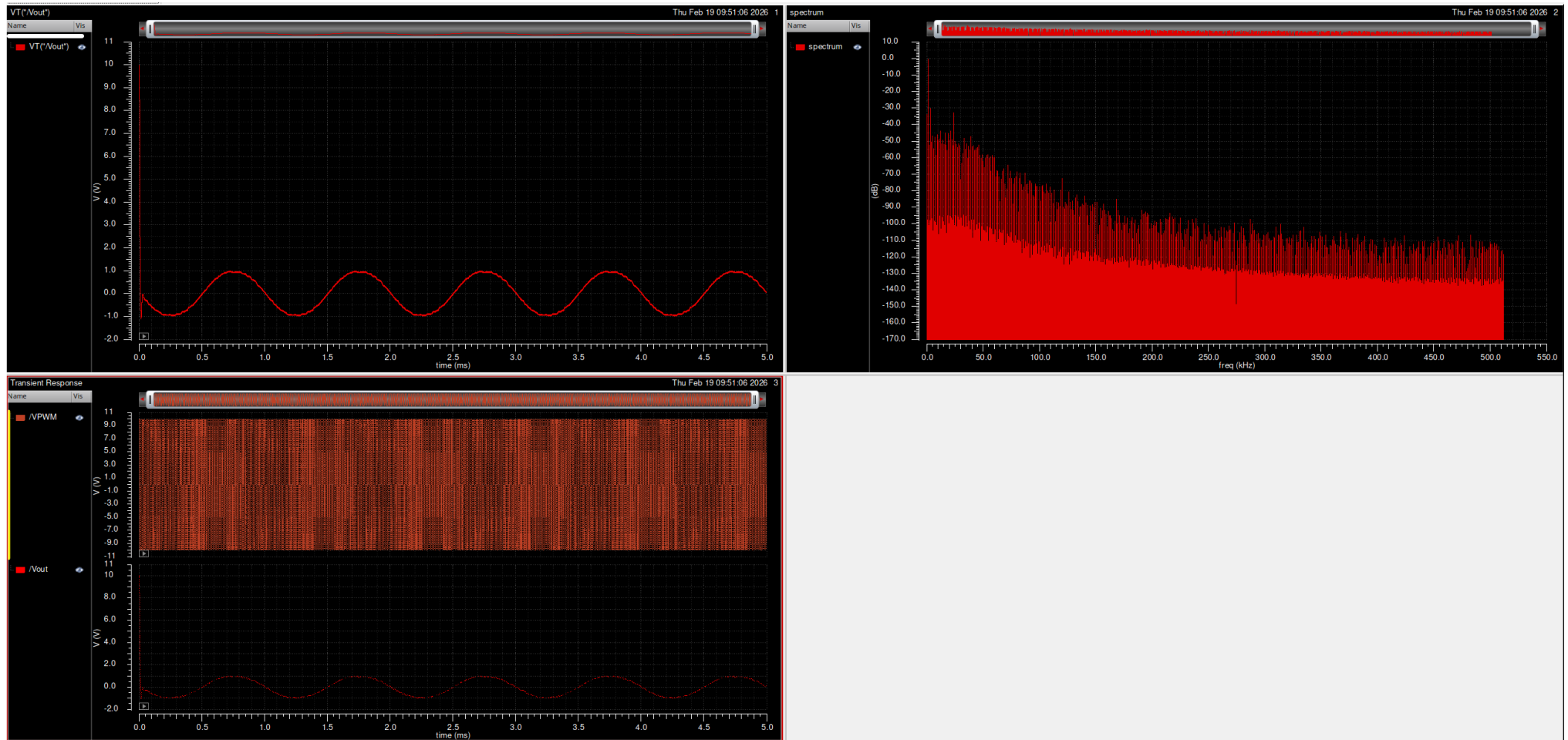


Part 2: 1.001 MHz, 0.1 FS



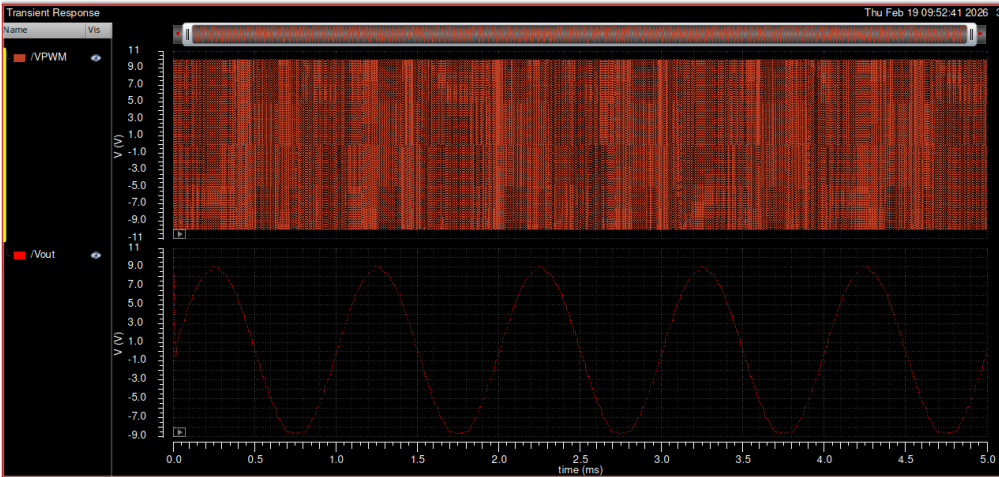
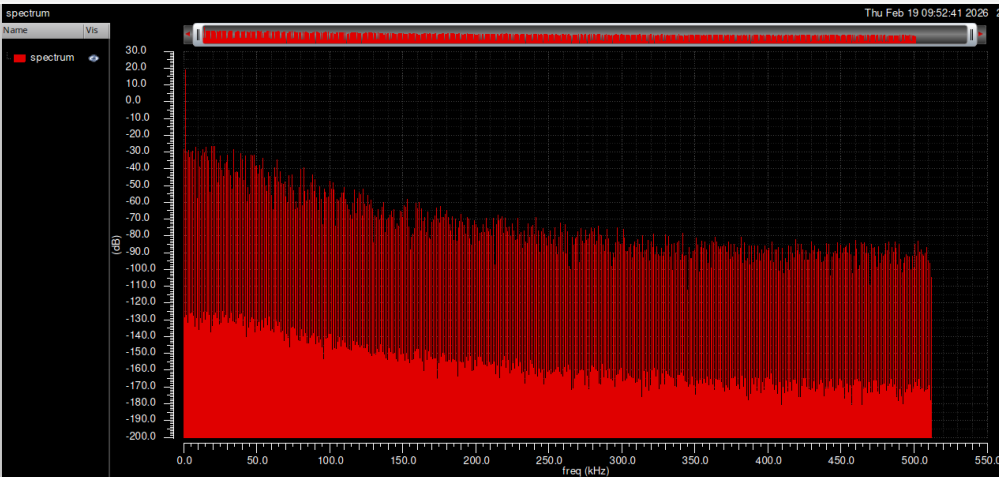
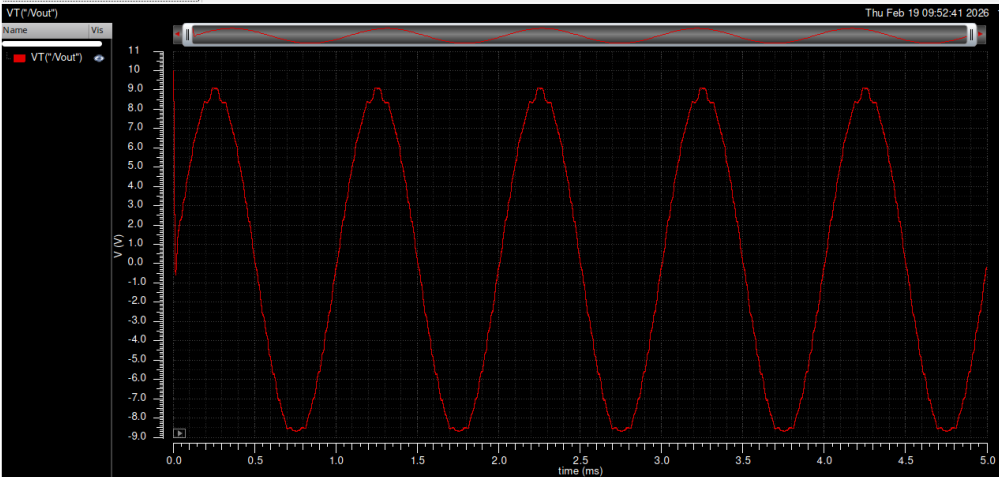
fPWM + 1kHz: attenuated by LC filter
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Part 2: 2.001 MHz, 0.1 FS



$2 \cdot f_{\text{PWM}} + 1 \text{ kHz}$: aliasing back to 1 kHz

Part 2: 1kHz, 0.9 FS



Part 3

Two-Phase Interleaved PWM

Part 3: Circuit & Theory

Two-phase interleaved Class-D:

Phase 1: comparator + L1 (18uH) + R1 (1m Ohm), carrier delay = 0

Phase 2: comparator + L0 (18uH) + R2 (1m Ohm), carrier delay = 500ns (T/2)

Shared output: C = 820nF || R = 4 Ohm

DC input: vdc = mi (modulation index, swept 0 to 0.9)

Single-phase ripple:

$$dl_{1ph} = 5 \cdot (1 - mi^2) / 18 \text{ [A]}$$

Two-phase combined ripple (D >= 0.5):

$$dl_{2ph} = 10 \cdot mi \cdot (1 - mi) / 18 \text{ [A]}$$

Reduction factor: $dl_{2ph} / dl_{1ph} = 2 \cdot mi / (1 + mi)$

At mi=0: perfect cancellation. At mi=1: no benefit.

Part 3: Ripple Current Comparison

mi	D	dl_1ph (mA)	dl_2ph (mA)	Reduction
0.0	0.500	278	0	100%
0.1	0.550	275	50	82%
0.2	0.600	267	89	67%
0.3	0.650	253	117	54%
0.4	0.700	233	133	43%
0.5	0.750	208	139	33%
0.6	0.800	178	133	25%
0.7	0.850	142	117	18%
0.8	0.900	100	89	11%
0.9	0.950	53	50	5%

Part 3: Key Observations

Single-phase ripple: max at $m_i=0$ (278 mA), decreases with m_i

Two-phase combined ripple: ZERO at $m_i=0$ (perfect cancellation)

Peaks at $m_i=0.5$ (139 mA), then decreases

Two-phase always $\leq$ single-phase ripple

Max single-phase (278 mA) = 2x max two-phase (139 mA)

Greatest benefit at low modulation (idle/low-power):

At $m_i=0$: 100% reduction

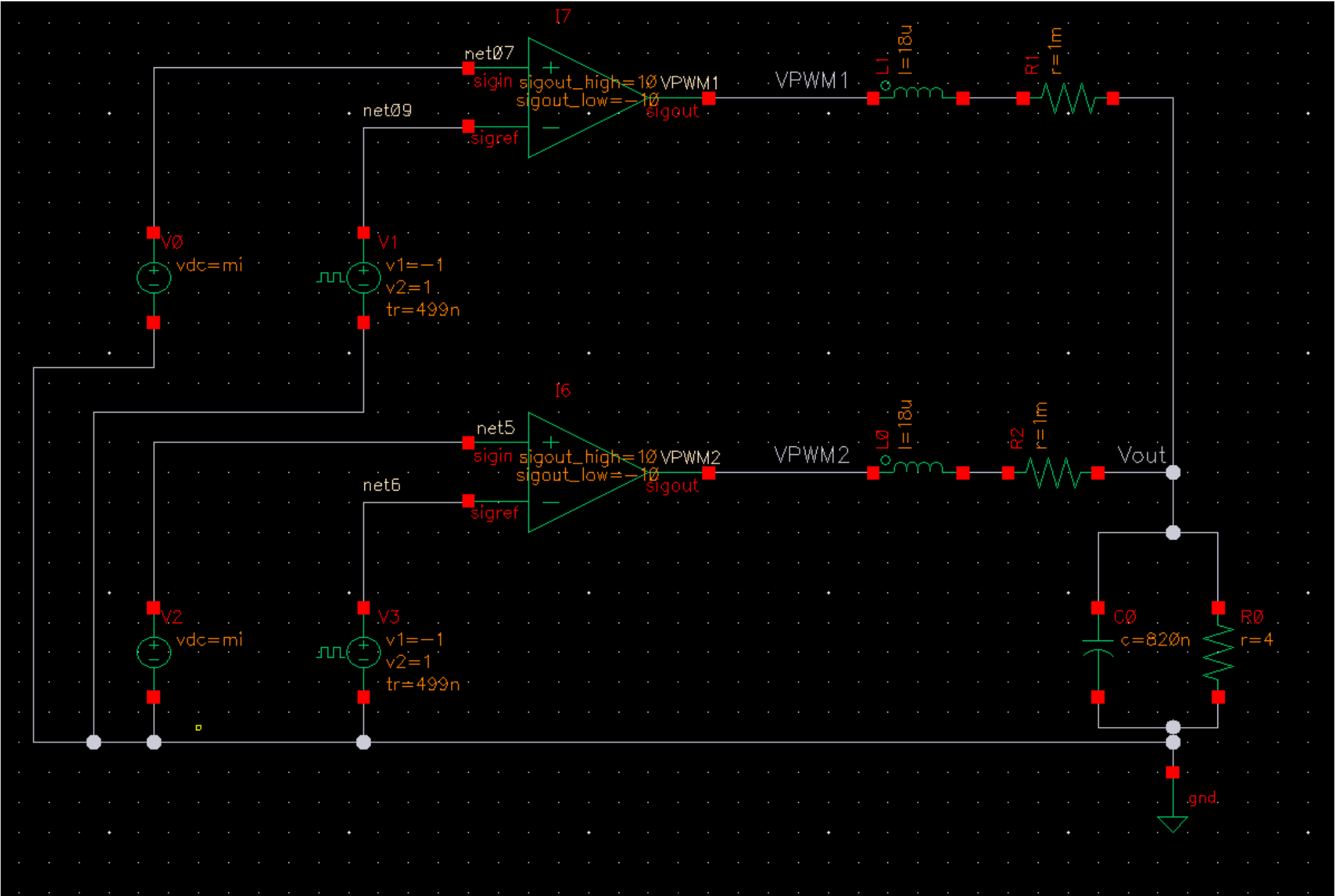
At $m_i=0.5$: 33% reduction

Simulation confirms analytical predictions:

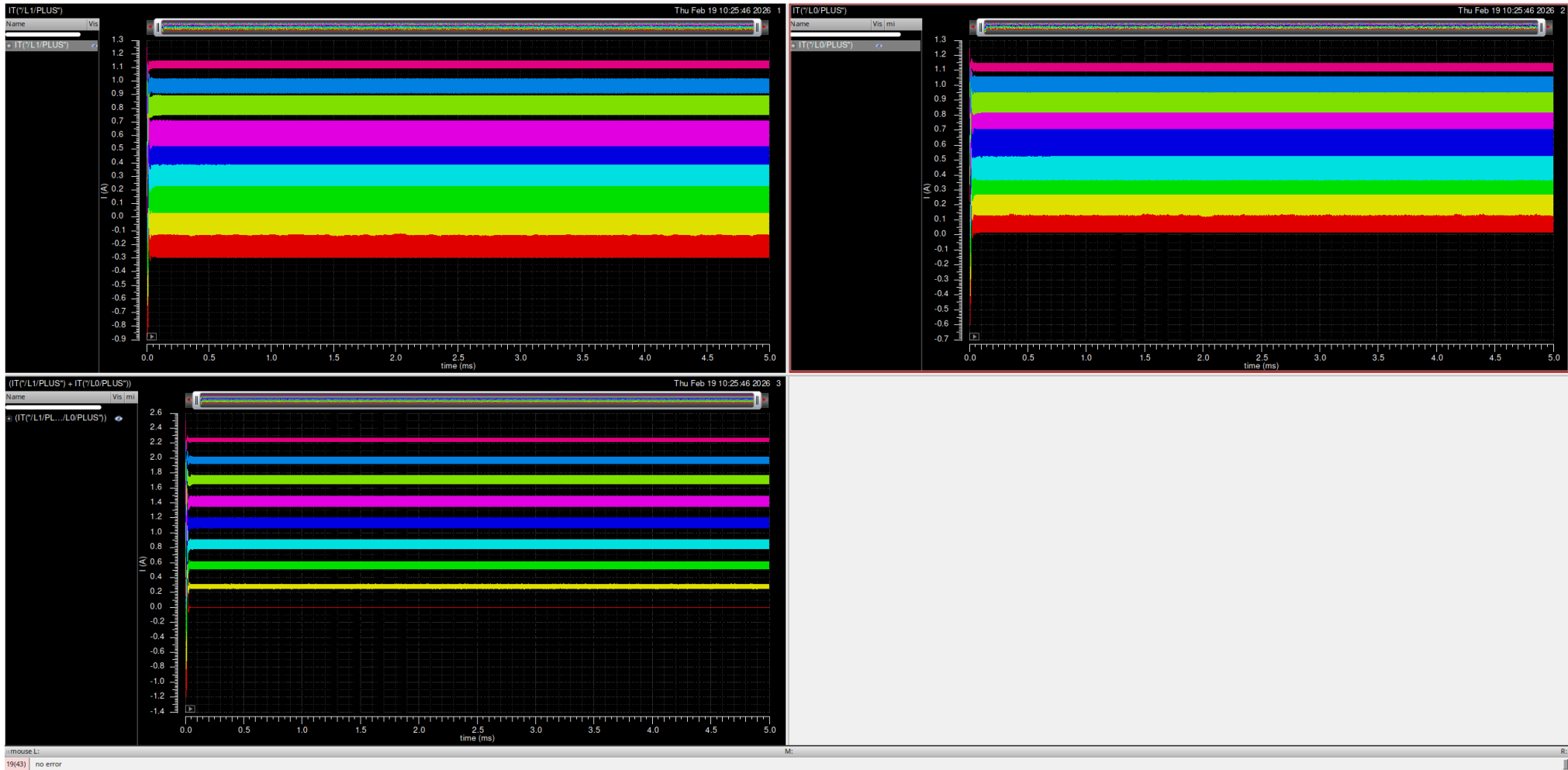
Combined current at $m_i=0$ shows near-zero ripple

Ripple increases with m_i as expected

Part 3: Schematic

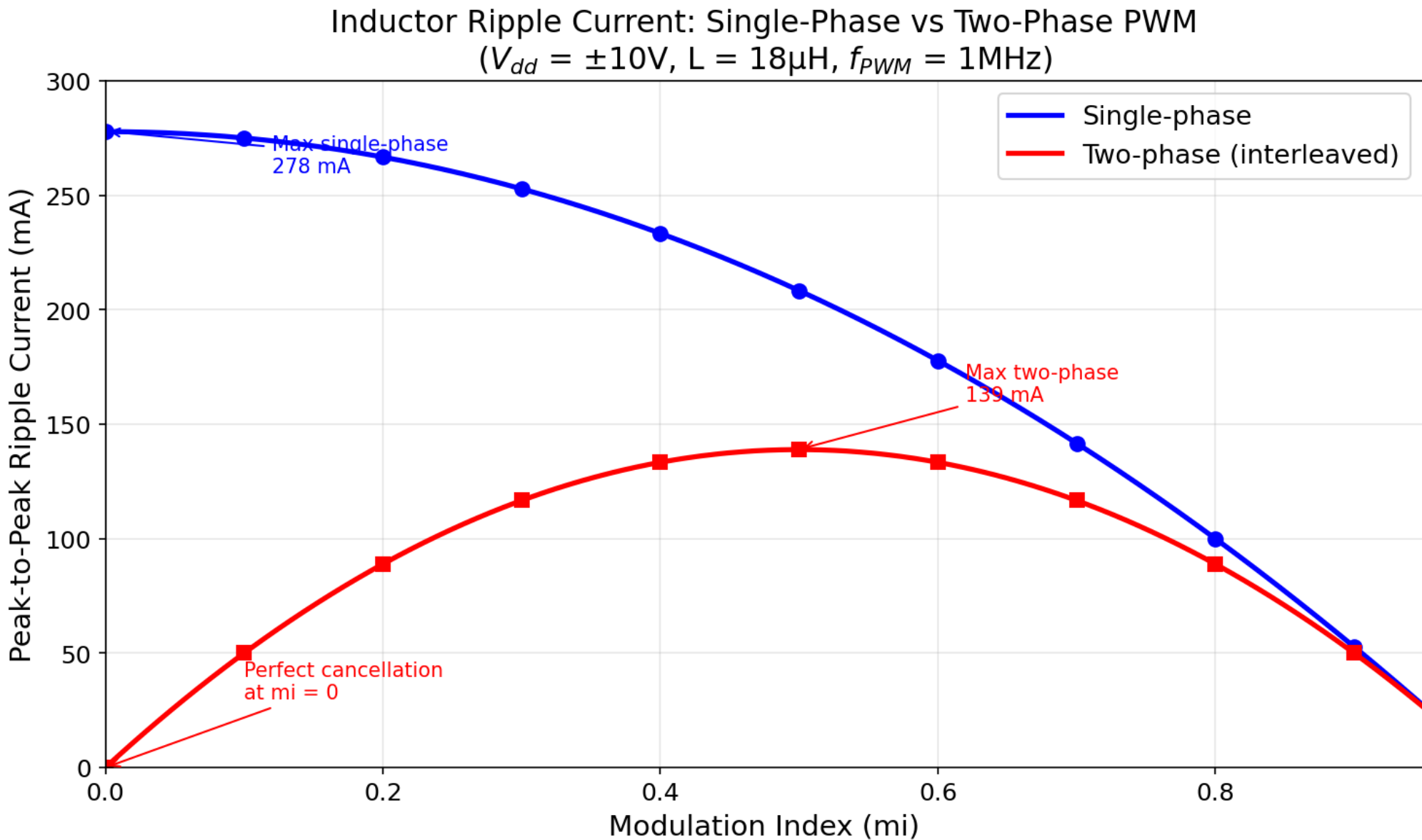


Part 3: Parametric Sweep Results



Inductor currents: Phase 1, Phase 2, and combined ($m_i = 0$ to 0.9)

Part 3: Ripple Current vs Modulation Index



Summary

Part 1: LC Lowpass Filter

$L = 18 \text{ uH}$, $C = 820 \text{ nF}$, $f_0 = 41.4 \text{ kHz}$, $Q = 0.854$

55.3 dB attenuation @ 1MHz (>50dB in all tolerance corners)

Part 2: Double-Sided PWM

Comparator-based PWM at 1MHz, +/-10V output

LC filter recovers audio signal, attenuates carrier

Aliasing at $2 \cdot f_{\text{PWM}}$ confirmed (key property of double-sided PWM)

Part 3: Two-Phase Interleaved PWM

180-degree phase shift between two PWM channels

Ripple cancellation: up to 100% at $m_i=0$

Two-phase max ripple (139 mA) = half of single-phase max (278 mA)